

PLECTA: Overlap-aware instance reconstruction of dense strand networks from binary masks

Oday Allan^a Stefan Luding^a Igor Ostanin^{a,*}

^a Multi-Scale Mechanics (MSM), Faculty of Engineering Technology, University of Twente, Enschede, The Netherlands

ORCID: Oday Allan [0000-0002-6341-4695](#); Stefan Luding [0000-0001-7598-0929](#); Igor Ostanin [0000-0003-4233-4255](#)

Abstract

Instance segmentation on a dense network of long, thin strands is difficult because it can fail in two opposite ways: a loose reconnect algorithm ends up fusing the wrong strands, and a strict one ends up fragmenting them, so that each strand is split into multiple pieces. A non-perfect binary mask, such as one from machine-learning segmentation, makes both failures harder to avoid, because it poses the same two ambiguities at every junction and every gap: a true crossing, where separate strands legitimately fuse into one connected component, cannot be told apart from an erroneous merge, and a true end cannot be told apart from a break left by a fading segmentation. Resolving either ambiguity takes the same judgement: which arm continues into which.

We present PLECTA, which makes that judgement from a binary strand-axis mask and nothing else. It cuts the skeleton into arms and junctions and solves every junction, of any degree, as a single exact maximum-weight *partial* matching in which leaving an arm unmatched carries an explicit price. A strand is therefore allowed to end, instead of having a continuation invented for it. Crossing pixels are emitted to every strand that claims them, so the output is a set of overlapping layers rather than the hard partition most instance methods return. The evidence is purely directional (tangent and curvature, read over tens of pixels either side of the discontinuity), which makes the working condition plain: the method applies where direction survives a crossing, as it does for nanotube bundles, biological filaments such as actin and collagen, and synthetic fibres, and not for freely flexible chains.

On 128 synthetic scenes from unseen seeds, PLECTA reaches mean pairwise $F_1 = 0.85$ [0.84, 0.86], falling from 0.91 to 0.77 as coverage rises from 20 to 60 %. It leads three released implementations, run from their own sources on identical masks, at every density. The comparison was repeated on a competitor’s own generator, GraFT’s, vendored unmodified at its published settings, over 40 scenes. The ordering held at every density stratum, at F_1 0.92 against 0.48 and 0.35.

A secondary depth-resolution stage then reads the greyscale to decide which strand lies on top at each crossing, and stacks the network into layers, an ordering otherwise obtained by destructive sectioning. It is correct on 0.88–0.96 of the crossings it commits to, against a chance level of 0.5, because it abstains where the image does not separate the two by more than the local noise. It abstains at 8–18 % of crossings, and counting every one of those as an error it is correct on 0.56–0.81 of all reference crossings. The method uses no random number generator and its configuration is published, so a mask yields the same strands on every run. We demonstrate the whole workflow on three annotated carbon-nanotube SEM fields, where the ordering survives on real masks, and state plainly what that evidence does not support.

Keywords: strand networks; instance segmentation; graph matching; geometric reconstruction; carbon nanotubes; scanning electron microscopy.

*Corresponding author: i.ostanin@utwente.nl

1 Introduction

Many materials are composed of long, thin strands laid over one another, some tangled at random and some deliberately twisted [1]: sheets of carbon nanotubes, collagen in connective tissue, paper, felt, nonwoven textiles, fibre-reinforced composites, aramid fibre ropes and synthetic fibre mats. The effective properties of such a material (tensile and flexural stiffness, electrical and thermal conductivity, fracture, tear and impact resistance) are governed as much by the architecture of the network as by the constituents themselves, and often more so: the stiffness and strength of a nanotube assembly are bounded by load transfer between tubes rather than by the very high modulus of any one of them [2–8]. The governing descriptors are therefore geometric: the length of each strand, its thickness, its orientation, its local curvature, and its connectivity to the strands around it. The same descriptors are the measurement target well outside materials science. A DNA fibre spreading assay reads the length ratio between two labelled replication tracts along an individual fibre [9], and time-resolved imaging of the actin cytoskeleton follows how its filaments reorganise inside a living cell [10].

A micrograph shows that network directly. Turning the picture into those numbers is the hard part. Routine image analysis measures the network in bulk (what fraction of the frame is covered, which way the texture points [11–13]) and stops there. To measure a distribution of strand lengths, or to ask how often one strand crosses another, something more is needed: every strand must be recovered as a separate object, addressable from one end to the other.

That is an instance-segmentation problem, and it has an awkward geometry. The objects are long, thin and open rather than compact and closed; they touch, cross and hide one another; and what tells two of them apart is not shape or size but which way each was heading. The consequence is a pair of failures that pull in opposite directions. Where two strands cross, their pixels join into a single connected blob, the same blob a genuine merge error would leave, and a method that trusts connectivity reports one object where there are two. Where the image fades or the segmentation drops out, one strand arrives in several disconnected pieces, the same gap a genuine end would leave, and the same method reports several objects where there is one. Both situations are repaired by answering one question: at this junction, which arm continues into which?

Existing tools answer part of it. Semantic segmentation is mature, since U-Net and its self-configuring descendants reliably separate strand foreground from background [14, 15], in materials imaging as much as anywhere [16, 17], and nanotube micrographs have their own trained segmenters [18–20]. But a foreground mask carries no identity, so it inherits both failures rather than resolving them. Instance methods assume the objects are compact, in materials microscopy [21] as in the general-purpose families: bounded blobs, centre-directed flows, star-convex outlines, none of which describes a long open curve that may overlap another [22–24]. Ridge and vesseness filters sharpen curvilinear signal but assign nothing across a junction [25–27], and the established fibre-analysis packages return network statistics rather than resolved objects [28–32]. Where the imaging is volumetric the problem is different in kind, because strands that appear to meet occupy disjoint volumes: in X-ray tomography of fibrous solids and of twisted ropes, individual fibres are separated by seeded watershed and morphological reasoning on the reconstructed volume [8, 33]. A projection has no such recourse, since two strands that cross genuinely share the same pixels. This paper addresses the step those leave open: given a binary strand-axis mask, recover the individual strands through the crossings and the gaps.

Recovering strands through crossings is well-trodden ground. Existing methods differ in two ways that matter: what evidence they read, and whether a pairing once made can later be undone. On both counts the field clusters.

The largest cluster follows one recipe. These methods skeletonise the mask, cut the crossing out, and pair the arms left behind by how nearly each continues the direction of another. Each

pairing is accepted as soon as it is scored. SIFNE reads its tip directions from the excised stubs and recovers dense microtubule networks from the mask alone [34]. Wegmayr et al. pair skeleton branches by lowest mutual angle [35]. GraFT covers a Frangi-filtered skeleton greedily, longest path first [10]. This is cheap, and where crossings are rare it is enough. Its weakness is structural: once an arm is paired, a later arm that contradicts the choice cannot reopen it.

Two ways out of that have been tried. The first is to bring in more evidence than a mask carries. Wegmayr et al.’s stronger arm drops the angle rule and learns pixel embeddings on patches around each junction; it reads the greyscale image, and it must be trained on crops in which every fibre is already labelled individually. Even then it recovers tangled fibres at about half the precision of isolated ones, in data where two thirds of fibres cross nothing at all. Methods that pair ends with an orientation network, detect fibres with a region-based detector, or trace them recurrently [36–38] need the same labelled training data. Reading a mask and nothing else avoids that cost, which is the trade this paper takes. DNAi keeps to the mask and simply looks harder, enumerating every pairing a junction allows from its endpoint angles; it does so exhaustively at degree three and four, and not at all above [9].

The second way out is to stop deciding one pairing at a time. Basu et al. and DeFiNe both weigh a whole neighbourhood at once, which is what a busy node needs. Basu et al. delete every branch point from a spanning tree over centreline points, then reassemble the pieces with an integer program that prices each join by how sharply it bends, repeating until the set of pieces settles; a join that a later fit contradicts can therefore still be undone [39]. DeFiNe covers an already-built weighted graph with the smoothest paths it can find, and lets two paths share an edge, so a crossing pixel can belong to both strands instead of being awarded to one [40]. Neither, though, starts from an axis mask: one wants intensities, the other a graph some earlier stage has already produced. Both also reassemble at most three pieces at a time, which limits how much of a shattered neighbourhood a single round can repair. Deciding broadly and starting from a mask have so far been alternatives.

One method has them both, and it comes closest to the position taken here. De et al. reason globally and do start from a mask. Branches and junction contacts become a graph, and labels spread across it so that a decision at one node constrains its neighbours rather than resting on that node’s own angles [41]. The labels have to start somewhere, however. In their images they start at anatomical roots, the stained cell bodies of neurons or the optic disc of a retina, and the method returns as many strands as there are roots. A tangle of open-ended strands offers no such starting point, and how many strands it holds is precisely the unknown. Reasoning globally from a mask alone, with no seed and no strand count supplied in advance, is where this paper begins. Four of these methods are measured against it directly in Section 3.5, on identical masks and through one scorer.

Skeleton graphs, junction excision, continuation angles, distance gates, combinatorial optimisation, iterative refitting and overlapping output are therefore all prior art, and none is claimed here. What is new is narrower, and it is a combination: from a binary mask alone, PLECTA solves each junction of *any* degree as one exact maximum-weight *partial* matching in which leaving an arm unmatched is an explicit option at a fixed price rather than a parity leftover, alternates that with globally gated gap matching, re-fits its tangent and curvature frames along the chains as they assemble, and emits overlapping strands. Each element exists somewhere above; no method there combines them. Degree-general exact matching with a priced refusal is what lets a strand genuinely end rather than have a continuation invented for it. The closest single prior method is that of Liu et al., which likewise reads a mask, reconnects across gaps and permits overlapping output, but pairs termini greedily and depends on a trained orientation network [36]. That claim carries a condition, worth stating as scope rather than leaving to be inferred. The only evidence PLECTA has at a crossing is directional: the angle between two outward tangents, the turns onto the chord between them, and the continuity of signed curvature, each read over tens of pixels of arclength. The working condition is therefore a property of

the image: direction must persist across the junction at that scale, or the continuation is not decidable from geometry at all. Bending stiffness is what most directly produces such persistence, which is why the semiflexible regime is the method’s home ground: nanotube bundles sit there, and so do the actin, collagen and synthetic-fibre networks the methods above were built for. Stiffness contributes to the condition rather than uniquely causing it, since tension, adhesion, confinement or bundling can hold a direction just as straight. Where the condition fails, as for a freely flexible chain whose tangent decorrelates within the junction, no continuation is more plausible than another, and no amount of matching recovers what the image no longer distinguishes.

We introduce PLECTA, after the Latin *plecta*, a braid. It decomposes a skeleton into arms, endpoint stubs and merged junction nodes, alternates exact per-node continuation matching with globally gated gap matching while tangent and curvature estimates expand from local arms to assembled chains, and returns projected 2-D strand layers in which crossing pixels may be shared. Every parameter is fixed in a published configuration and the implementation draws on no random number generator, so a given mask yields the same strands on every run; that is the only sense in which it is called deterministic here.

Reaching real micrographs takes one further component, and the workflow is deliberately divided in two. An upstream route turns a raw SEM micrograph into a binary strand-axis mask by a synthetic-data-driven CycleGAN/U-Net whose training pairs are synthesised rather than annotated [42]; PLECTA is everything downstream of that mask. The division is what makes the grouping measurable on its own, against references whose identities are known exactly, and it keeps width rendering from moving an identity score. The closest complete workflow is that of Peters et al., who join a U-Net-like segmentation trained on over a thousand annotated SEM micrographs to algorithms that trace fibres through intersections, and validate against human evaluators on fibre number, length and width [43]. That work and this one divide the task at the same joint and differ in what the instance step is asked to prove: their tracer stands in for human counting and measurement, while PLECTA is isolated behind a binary mask and scored on identity through crossings against overlap-aware references in which every pairing is known. We evaluate PLECTA on 128 procedurally generated scenes spanning 20 to 60 % areal coverage. A further 50 scenes are rendered twice, once from a clean axis and once from a degraded one, so that what degradation costs is measured on unchanged geometry. Both stages are then demonstrated together on 3 annotated CNT-sheet SEM fields.

Beyond the plane, a crossing carries a second question the grouping does not answer: which strand lies on top. For nanostructured networks that is normally obtained destructively, by focused-ion-beam sectioning and tomographic reconstruction [44], because a micrograph’s intensities record surface topography rather than height. PLECTA answers it non-destructively from the greyscale already in hand. The machinery is deliberately conventional: monocular depth ordering has long derived pairwise depth from local occlusion evidence, assembled those relations into a depth-order graph and corrected the contradictions among them [45, 46], and single-view hair modelling resolves the layering of intersecting strands in much the same spirit [47]. What differs is the evidence available. Those methods read the shape of the outlines where two objects overlap, which a bundle a few pixels across in a noisy micrograph does not present reliably. What separates the two strands here is instead the continuity of each one’s own brightness: the upper strand keeps the appearance it has on either side of the crossing, while the lower strand’s is overwritten by the strand above it. The stage abstains where that difference is no larger than the local image noise. To our knowledge this is the first depth decomposition of a strand network from a single micrograph; it is reported as a demonstration with a measured accuracy, not as a validated three-dimensional reconstruction.

2 Materials and methods

This section fixes what the method is given, what it returns, and how both are measured: the input mask and the workflow around it, the synthetic and real data with the split that keeps them honest, the reconstruction itself stage by stage, the two stages that read the greyscale, and the measures every reported number goes through.

2.1 Scope and workflow

The input is a binary strand-axis mask $M \in \{0, 1\}^{H \times W}$, whose foreground $F = \{x : M(x) = 1\}$ is one pixel wide and traces strand centrelines rather than strand widths. From it PLECTA reconstructs an overlap-aware collection $\{I_1, \dots, I_N\}$ with $I_n \subseteq F$, so that every emitted pixel is one the mask already carried, and a crossing pixel may belong to several strands at once. The evaluated grouping stage uses only M ; where a greyscale SEM image exists it is read downstream, for width and brightness, and cannot revise the fixed grouping.

Figure 1 places the method. Its five geometric steps read mask geometry alone, and the two matching steps are re-made over eight rounds against frames re-fitted along the chains assembled so far. Where the mask itself comes from is a separate question, taken up in Section 3.3.

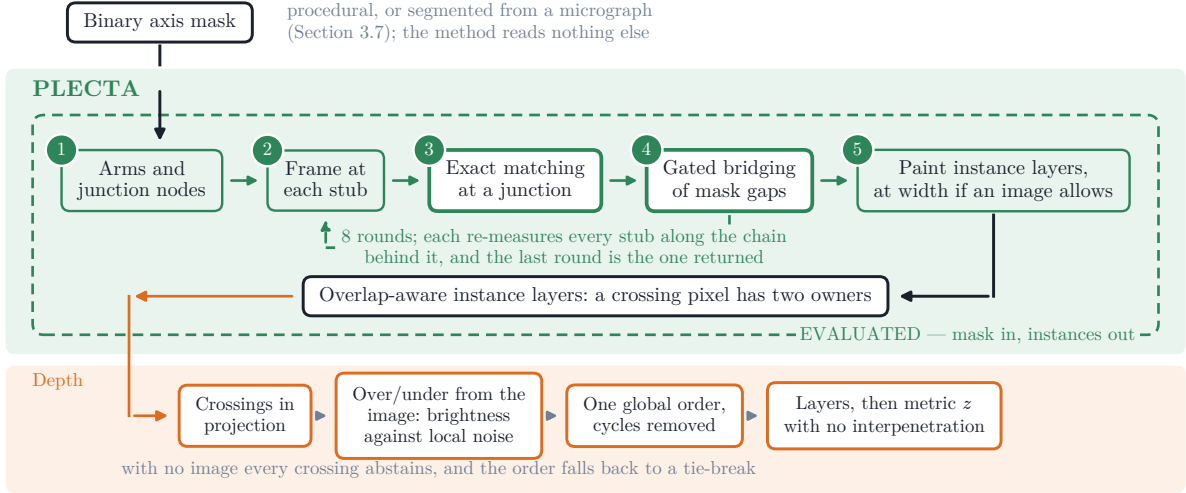


Figure 1. Where PLECTA sits. A single binary axis mask is the method’s only input, and every result in this paper measures the dashed green box. Steps 3 and 4 are the only decisions it makes and are drawn heavier for it; the rest is bookkeeping. The depth stage adds a third dimension without revising any identity. Colours: green, a PLECTA geometric step; orange, a dependency on the greyscale image; dark, a data object.

What makes the task hard is a single crossing: the mask holds one connected blob from which several arms leave, and nothing local states which arm continues into which. The output is a set of layers rather than a partition, so a pixel inside that blob is counted in both strands that claim it.

2.2 Synthetic data and evaluation split

Because the scenes are generated rather than annotated, the answer is known exactly. For every strand k the generator records its ordered centreline C_k and the pixels it was drawn onto, its support Ω_k . The supports are kept one per strand instead of being merged, so a crossing pixel belongs to both strands that pass through it and the reference is overlap-aware by construction.

Each scene is written out four ways: the exact one-pixel axis; that axis degraded with small gaps and raster irregularities; the strands at full width, merged into a single foreground; and an SEM-like greyscale image, in which each strand carries its own brightness and a slow variation along its length, blurred by one point-spread function over a low-frequency background with grain. A depth-resolved variant of the generator supplies the missing structure for the depth stage of Section 2.5.2. It gives every strand a height and a radius, refuses to let two interpenetrate, and renders the stack from the top down, so the image carries occlusion, defocus that grows with distance from a randomised focal plane, and width broadening. Those cues are randomised independently of one another, so no single rule such as brighter-is-higher recovers the order.

Scenes are labelled throughout by areal coverage, the fraction of the frame the full-width strands cover rather than the fraction the thin axis occupies. That label reports difficulty without setting it: drawing the same strands thicker raises coverage and leaves the score exactly where it was, since neither PLECTA nor the metric ever sees how wide a strand was drawn. What makes a scene hard is the amount of centreline packed into the frame, which coverage happens to follow under this generator, and Section 3.1 separates the two.

Every parameter was fixed on a separate development set and never on the scenes reported here; the 128 evaluation scenes, 32 at each of 20, 30, 40 and 60 % coverage, use seeds that appear nowhere in that development set. Supplementary Section S6 gives the seed blocks, the verifier that checks their disjointness, and the locked set that remains unevaluated.

A separate controlled paired set of 50 scenes, ten at each of 20, 30, 40, 50, and 60% coverage, was scored once from each geometry’s clean axis and once from its degraded mask, after the configuration was fixed.

2.3 CNT SEM data and mask sources

Three manually annotated CNT-sheet SEM fields were available, each 1536×1024 px: B58_100 (307 visible CNT bundles), B58_110 (430) and B58_300 (325). The annotated strand is the visible bundle, not the individual nanotube, which SEM does not resolve at this magnification. The nominal horizontal field width is $1.66 \mu\text{m}$ across the long, 1536-px axis of the stored images, which is 1.08 nm px^{-1} . Each annotation is one human interpretation of projected, locally ambiguous crossings.

The reference annotations were drawn with an in-house interactive instance-segmentation tool for greyscale SEM of thin, crossing strands, whose data model is per-instance masks rather than a flat label image, so a pixel may belong to several strands at once; that is what makes the reference overlap-aware and comparable with PLECTA’s output. The file the tool saves is the file the evaluation reads, with no export or conversion step in between. The annotator paints each visible bundle as its own strand, and the tool assists without deciding: strokes may snap onto the local intensity ridge, with polarity calibrated from the existing annotation, and may be re-rendered through a smoothing-spline fit whose centreline is shifted onto the measured intensity crest only where that crest is consistent; widths come from perpendicular full-width-at-half-maximum profiles referenced to the local minima either side of the peak and reported as twice the nearer half-width, so a crossing strand fusing into the profile does not bias the measurement wide. Candidate gap bridges, collinear endpoint pairs of different strands, are suggested but never applied automatically; each is accepted or rejected by the annotator, so identity through every crossing is a human decision, machine-assisted but not machine-made.

The protocol’s limits are stated rather than implied. The reference annotations are annotator A’s, and A is an author of this paper who also configured PLECTA. There was no blinding, so the reference may encode the method’s own reading of an ambiguous crossing, and no real-data difference smaller than the variation between two annotators should be read as real. These are manual reference annotations, not ground truth. Annotator B has annotated 3 of these fields independently, and Section 4 reports what that measurement shows: the two annotators do not implement one strand definition, so every real-data score in this paper is

reported against a named annotator and the two are never averaged. Scores throughout are computed on centreline fragments, not on strand widths.

Two inputs were derived per field. The first is the skeleton of annotator A’s strand annotation, an oracle-like condition that asks what the grouping can do when the axis evidence is curated. The second is an automatic mask from the upstream CycleGAN/nnU-Net segmentation of Section 3.3 [15, 42], which asks what an automatic mask allows it to do. That segmentation is a separate contribution with its own evaluation; all that matters here is that its output is a binary mask produced without seeing any annotation.

Mask quality was compared after skeletonisation with the symmetric 2-px tolerance used throughout, which avoids rewarding mask thickness, and reconstruction scores were computed within each mask’s own visible common-fragment population.

The fields were acquired on a Thermo Fisher Helios 5 UX DualBeam with an Elstar field-emission column in immersion mode, imaged with the through-lens detector at 2.00 kV and 400 pA and a working distance of 4.3 mm. Each frame is an integration of 35 passes at 100 ns dwell with drift correction enabled.

2.4 PLECTA reconstruction

The method proper. It reads the binary axis mask and nothing else, and the subsections follow the order it runs in: the graph it cuts the mask into, the cost that prices one join, the junction solve, the gap bridging, and the schedule that repeats the two as chain-level evidence accumulates.

2.4.1 Skeleton graph and stub frames

PLECTA skeletonises the mask, by Zhang and Suen’s thinning [48], and prunes the short spurs skeletonisation leaves behind. Skeleton pixels where three or more paths meet form junction clusters, and clusters close enough together are merged into a single *node*: a crossing seen at this resolution is one blob, not a point, and neighbouring crossings often touch. What remains between nodes is an *arm*, an unambiguous run of strand, carrying a *stub* at each end.

Each stub is given a frame: its tip, the direction the strand is heading as it leaves, and how it is curving. Both are fitted over a short run of arclength back from the tip, and a stub too short to yield a reliable direction is marked as such rather than guessed at. That support widens once arms are linked into chains, so a stub with no direction of its own borrows one from the chain it has joined, which is why the method iterates rather than deciding once.

2.4.2 The pairing cost

Linking two stubs is priced by asking three geometric questions (Figure 2). Do the two directions oppose one another, as they would if one strand simply continued into the other? Does each direction point at the other’s tip, rather than running parallel past it? And does the curvature continue, or would the join require a kink? For a gap the distance to be spanned is charged as well; at a junction there is no gap to pay for. The cost of linking stubs a and b is therefore

$$C_{ab} = w_d \theta + w_t q(d) (\varphi_a + \varphi_b) + w_\ell \frac{d}{\ell_0} + 30 w_\kappa |\kappa_a + \kappa_b|, \quad q(d) = \min\left(1, \frac{d}{d_0}\right), \quad (1)$$

with d the tip separation, θ the reversal angle between the two outward tangents, and φ_a, φ_b the turns from each tangent onto the tip-to-tip chord, measured at their own vertices; junction links set $w_\ell = 0$.

Three properties of that expression are deliberate. The reversal angle ignores the chord, which keeps it meaningful when the tips are three pixels apart and the chord direction is quantisation noise. The chord turns do see lateral offset, so two parallel arms that never meet are penalised.

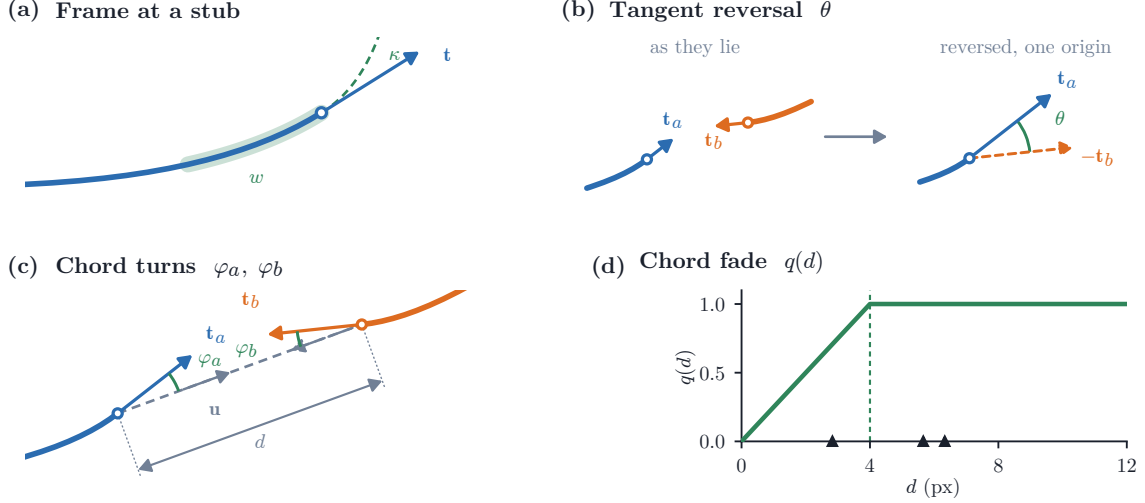


Figure 2. The stub frame and the terms of the pairing cost. **(a)** The least-squares arclength fit over window w , read outward from the tip at $s = 0$, giving outward tangent $\mathbf{t}$ and signed curvature κ ; quadratic when the span allows, linear otherwise. **(b)** θ , in the two steps its definition takes: the outward tangents where they lie, then the same pair with $\mathbf{t}_b$ reversed and both read from one origin. A perfect continuation makes them antiparallel, so $\theta = 0$. **(c)** The chord turns φ_a, φ_b onto the chord $\mathbf{u}$ joining the tips. **(d)** The fade $q(d)$ against tip separation; triangles mark the six real separations at the crossing of Figure 3.

And because it is the chord terms that degrade over short separations, the fade $q(d)$ withdraws them as the tips approach rather than letting an unstable angle dominate. The weights, the scales ℓ_0 and d_0 , and the constant placing curvature on the scale of the angular terms are all fixed in the released configuration and are listed in Supplementary Section S1.

2.4.3 Candidate gates and exact matching

Leaving a stub unmatched is priced rather than forbidden, and that price does double duty: it also fixes what may be offered at all, since an edge costing more than the two prices it would save could never be chosen. Exact maximum-weight partial matching then picks, over disjoint pairs,

$$\max \sum_{(i,j) \in \mathcal{M}} (p_i + p_j - C_{ij}), \quad \text{admissible iff } C_{ij} < p_i + p_j, \quad (2)$$

which is the same thing as minimising the total link cost plus a price p_i for every stub left free. Junction and gap links carry different prices, and a gap link must additionally clear limits on how far it may reach and how far either direction may turn, the tighter set, because a gap asserts a continuation across evidence that is *missing* rather than merely ambiguous. Supplementary Section S1 lists the prices, the limits, and the schedule below.

The method does not decide once. It runs the two matchings over several rounds, re-fitting every frame along the chains assembled so far before each. Early rounds are deliberately conservative: prices are scaled down, which admits fewer edges, and the gap angle limits are narrowed, so only pairings that survive the stricter test are committed while the frames are least reliable. The constraints relax towards their configured values as the chains sharpen the frames, and only the final round runs at the configured parameters. That round is the one returned.

Every stub is therefore either paired or pays the unmatched price, and there is no third possibility. That priced option is what lets a strand genuinely end, and what stops a node with an odd number of arms from inventing a continuation for the one left over. No node is classified: a node of any degree is solved by the same matching, and an arm ends because leaving it unmatched won the objective, not because the node was recognised as a particular kind.

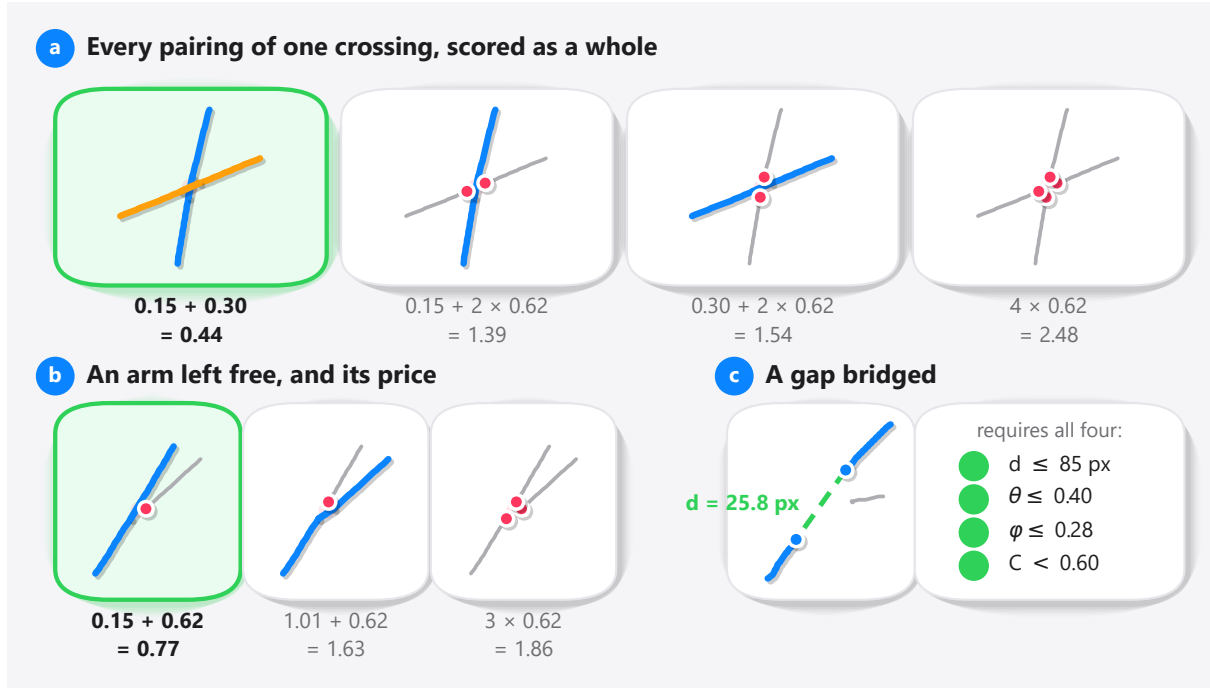


Figure 3. Every decision the linker makes, on the nodes it made them at. Every arm is the skeleton the numbers were measured from, so every angle shown is the one the cost was computed on. **(a)** Every pairing the admissibility gate allows at a four-arm crossing, each scored as a whole: its edge costs plus the price $p_i = 0.62$ of each arm it leaves free. The green card is the one returned; two further pairs exceed $p_i + p_j = 1.24$ and are never offered. **(b)** A three-arm node where an admissible edge is declined on the total rather than on its own cost. **(c)** A gap link, which must clear all four gates at once.

It also removes a parameter rather than adding one. Admissibility in Equation 2 is not a threshold chosen alongside the price: an edge costing $C_{ij} \geq p_i + p_j$ contributes nothing positive to the objective, so a maximum-weight *partial* matching could not select it under any setting, and declining to offer it removes only edges that could never have been chosen. One number therefore states both what a refusal is worth and what may be considered, where a method that gates separately must set a cost threshold and a price and keep them consistent.

Solving a junction as one matching rather than one edge at a time is the distinction claimed in Section 1, and Figure 3 shows what it buys on real nodes, alongside the other two decisions the linker makes. A locally attractive edge can be declined, as at the three-arm node of Figure 3b. Joint and sequential solutions coincide at plain crossings and diverge only as nodes grow dense: they never disagree at nodes of degree two to four, disagree at roughly a fifth of nodes with five to eight stubs, and at two thirds of nodes with nine or more, where the joint solution is the cheaper one; Supplementary Section S1 gives the counts. That comparison holds one node's costs fixed. Run inside the rounds, where each matching sets the frames the next is scored against, a cheap early choice propagates and the difference is no longer confined to dense nodes.

Two measurements test that claim rather than assert it. The greedy continuation baseline of Section 3.4 reads the *same* masks and the same local geometry and differs only in pairing greedily instead of jointly, so the $+0.145$ F_1 between them isolates the formulation rather than the evidence. The component ablations reported alongside it then ask the complementary question, whether the rest of the combination is load-bearing or decorative, and find that removing the junction chord-turn term costs -0.1858 and removing gap matching -0.1394 . Neither measurement alone would do: the first says the matching is worth its complexity, the second that the terms it matches on are not redundant.

2.4.4 Strand output

Connected components of accepted arm links define strands. PLECTA paints every arm pixel and every junction cluster touched by a chain into that strand, so intersecting strands share crossing pixels. An isolated one-arm component shorter than 6 px is omitted. Accepted gap links determine identity but are not painted into the fixed centreline output. All effective parameters, including values previously inherited as code defaults, are materialised in the fixed JSON configuration.

2.5 Image-informed reconstruction

The grouping above reads the mask alone and is what this paper evaluates. Where a greyscale image of the same field exists, three further stages can read it: two give each strand a width, and one orders the strands at their crossings. All three run after the grouping is fixed and none can move a pixel from one strand to another, so no identity score in this paper depends on any of them. All are governed by limits recorded in the fixed configuration, and none is specific to SEM: they take whatever image accompanies the mask, and SEM is simply the case demonstrated here.

2.5.1 Width and ribbon rendering

Perpendicular intensity profiles, sampled along each ordered chain away from junctions, attach an apparent width, a brightness, an uncertainty and a quality flag to an existing strand without altering its arm membership. Ribbon rendering then interpolates accepted gaps, resamples and smooths the centreline, and draws the chain at its fitted width, excluding junction blobs. Neither silhouette absorption, which would union rendered strands, nor any other step merges two strands after the fact, and none contributed to the held-out result.

2.5.2 Depth decomposition

A projected crossing has an owner in each of two strands but no order between them. This stage supplies one.

Evidence at one crossing (Figure 4). Where two strand centrelines intersect, a short arc of each is taken through the crossing (the shared *core*, a fixed length, the same at every crossing), together with the *flanks* on either side of it, kept clear of the other strand's stroke so that they measure one strand and not the union. One channel is read on both: median intensity. The physical argument is occlusion. Whichever strand lies on top presents the same surface through the crossing as it does beside it, so its core matches its own flanks in brightness; the lower strand is overwritten there, so its own continuity breaks. The strand whose flank brightness the core matches better is called upper, and by how much, divided by the larger of the two strands' brightness separation and the noise measured at that crossing, is the score s that decides it. Where the two strands genuinely look alike, or differ by less than that noise, $|s|$ falls below a fixed threshold and the crossing *abstains* rather than being decided by noise. Dividing by the noise, rather than by the separation alone, is what makes a confident score mean *separated by more than the noise* rather than merely separated.

One order for the whole field. Local decisions need not be mutually consistent: they can assert that A is above B , B above C and C above A , which no assignment of heights satisfies. Each decided crossing therefore contributes a directed edge weighted by $|s|$, the strength of its own evidence (Figure 5), and within each connected component of the crossing graph one total order over its strands is derived so as to maximise the total satisfied weight, which is the same as reversing the lightest set of relations that leaves no contradiction behind. That is the minimum-weight feedback arc set objective, and it is attained exactly only for components small enough to search, larger ones being approached greedily. Every relation is then re-read from

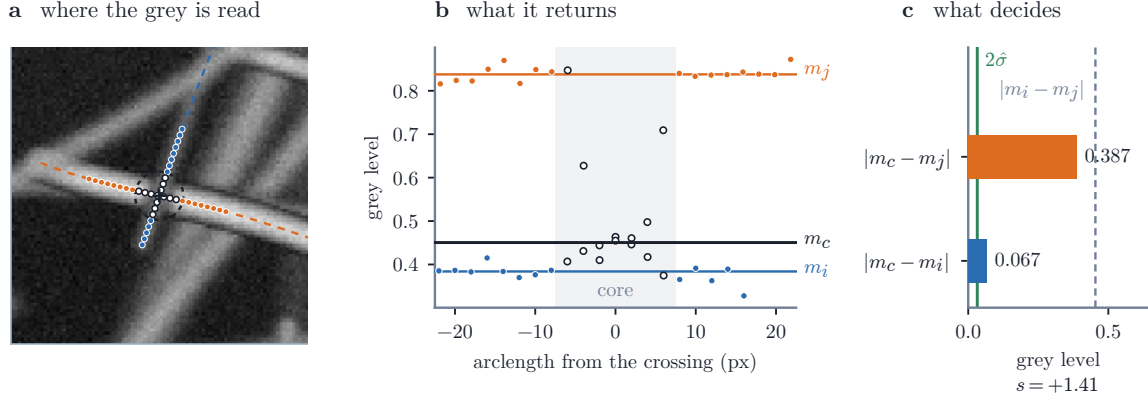


Figure 4. What the depth stage measures, at one crossing where two strands meet at 83.8° , chosen as a legible example rather than at random. Every position and grey level was read back from the stage itself. (a) The core arc shared by the two strands, and each strand’s own flanks, which the window keeps clear of the other’s stroke; open dots are core samples, filled ones flanks. (b) The same samples against arclength: the pooled core median m_c sits by strand i ’s flank level and far from j ’s, which is the occlusion. (c) The two distances the score compares, over the larger of the strands’ contrast and twice the flank noise. That ratio is the score s of Supplementary Section S2, whose sign names the upper strand and whose size weights the crossing below.

that order, which is re-derived for the component as a whole rather than only where relations conflict, so a relation that nothing contradicts can come back reversed too; the relations the order disagrees with are flipped rather than discarded, and both the flip and the original local score are recorded, so a reader can see how far the global correction moved. Depth propagates only along crossings, so two strands in different components are related by no evidence at all and their order is a stated tie-break, not a measurement. Supplementary Section S2 gives the sampling geometry, the score in full, the measurement that replaced an earlier two-channel rule with it, and the solver.

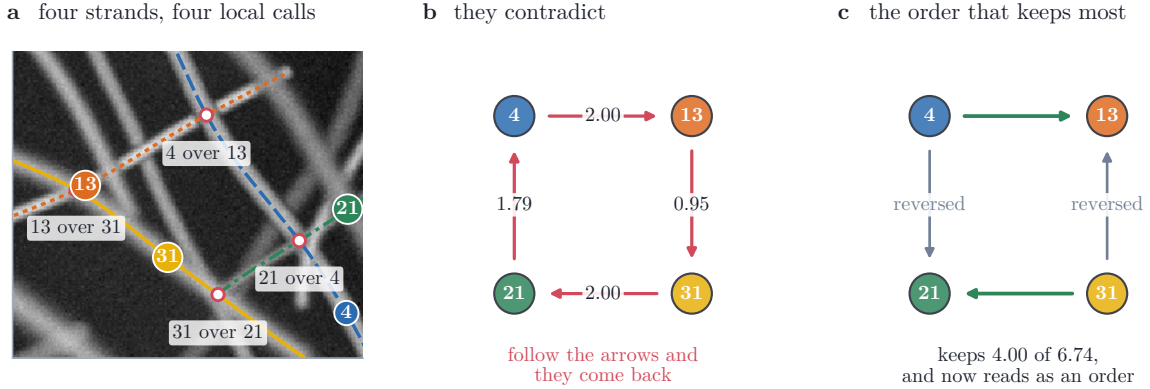


Figure 5. Why the order is solved for a whole component rather than one crossing at a time, on a contradiction the stage met. (a) Four strands and the four calls the local evidence made at their crossings. (b) Those calls as a graph: each arrow points from the strand called upper to the one called lower, and following them returns to where it started, so no stack of heights satisfies all four. (c) The order satisfying the most evidence by weight keeps the two heaviest calls and reverses the two lightest, which are recorded rather than discarded. Layer numbers are not shown: a strand’s layer follows from its whole component, and these four sit in a scene of seven.

Layers and height. The corrected relations form a directed acyclic graph whose longest path fixes the smallest number of layers consistent with them; strands that never conflict share a layer, so the layer count follows from the ordering rather than from the number of strands. Each

strand is then placed at a height that respects its order and keeps strands from interpenetrating at their measured radii, with the stack compacted so that the film is as thin as those constraints allow.

Without a greyscale image every crossing abstains, the ordering has nothing to satisfy, and the output degenerates to the tie-break: the stage requires image evidence and reports how much of it was usable. Section 3.2 evaluates it, and Supplementary Section S5 defines the measures used, which are disjoint from those above because none of them can see depth.

2.6 Metrics and statistical analysis

The primary endpoint is common-fragment pairwise F_1 . The input mask is cut into fragments, each one unambiguous arm between crossings, by a rule that reads neither the reference nor any prediction, so every method compared is cut into the same pieces; each fragment then takes a reference and a predicted identity by one rule applied to both sides alike, and the score counts the pairs of fragments the prediction groups as the reference does. Figure 6 works that endpoint through the smallest case that has a crossing, including the two ways it can be undefined rather than zero.

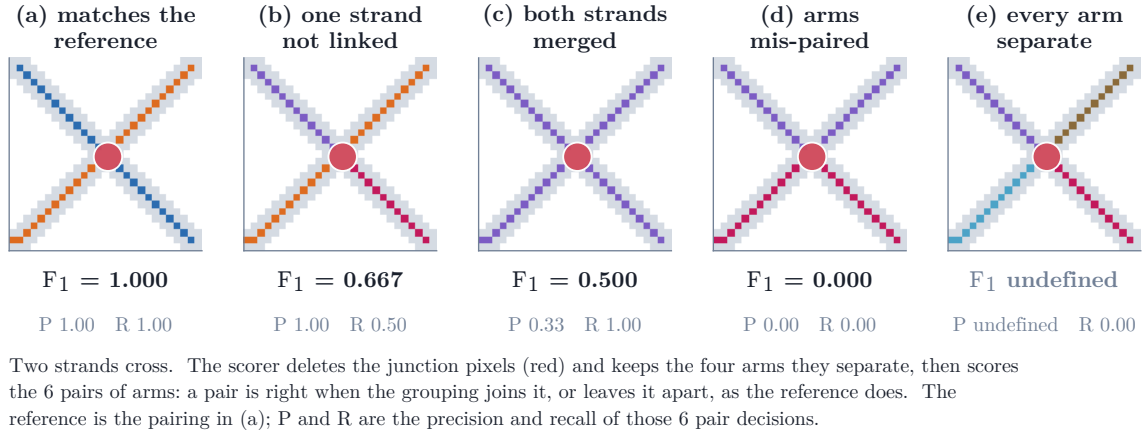


Figure 6. What the endpoint scores, on the smallest case that has a crossing. Two strands cross; the scorer removes a neighbourhood of the junction (red) and keeps the four arms it separates, then asks of each of the six pairs of arms whether the grouping joins it, or leaves it apart, as the reference does. (a) is the reference pairing. (b) leaves one strand unlinked, a split, which costs recall alone. (c) merges both strands into one instance, which costs precision alone. (d) pairs the arms the wrong way round and scores zero, having got no pair right. (e) emits every arm separately, which makes no pair at all, so precision has no denominator and F_1 is undefined rather than zero, which is one reason the adjusted Rand index and variation of information are reported beside it. Every value is computed by the scorer of Supplementary Section S3, not worked out by hand.

Supplementary Section S3 gives the construction and the assignment rule in full. Because junction pixels are deleted before fragments are labelled, the measure cannot see whether an emitted crossing pixel is given one owner or two, only whether the arms meeting there are grouped as the reference groups them.

No one measure is quoted alone, because each is blind to something another sees: pairwise F_1 scores relationships between pieces and is quadratic in fragmentation, while object detection scores existence and cannot see an arm swap at a crossing, which it rates 1.000 where pairwise scores 0.000. The adjusted Rand index and variation of information [49, 50] are therefore quoted beside the pairwise score, with detection F_1 and Crossing Fidelity. The input masks themselves are characterised by a tolerant recall, precision and Jaccard index, called completeness, correctness and quality in the thin-structure literature, matched within a tolerance rather than pixel for

Table 1. Held-out reconstruction scores overall and by target areal coverage. VI denotes variation of information, for which lower is better; higher is better for the other metrics.

Group	n	F_1	Precision	Recall	ARI	VI split	VI merge	VI total
All	128	0.85	0.87	0.84	0.85	0.26	0.22	0.47
20%	32	0.91	0.93	0.90	0.91	0.14	0.09	0.23
30%	32	0.87	0.89	0.86	0.87	0.21	0.16	0.37
40%	32	0.86	0.86	0.85	0.85	0.24	0.22	0.47
60%	32	0.77	0.79	0.76	0.77	0.44	0.39	0.82

pixel, because two skeletonisations of one strand seldom fall on the same pixels. Supplementary Section S3 defines each in full, with its equation, what it rewards and what it cannot see.

Supplementary Section S3 states the interval construction, the resampling unit and the cases where no inference is drawn.

3 Results

The results run in the order the evidence allows. PLECTA is measured first where the answer is known exactly, then the depth stage that reads the greyscale on top of it, then both on real micrographs; the last two subsections say what each part of the method contributes and how it stands against published alternatives.

3.1 Reconstruction on synthetic images

PLECTA is scored first where the answer is known exactly: the held-out synthetic scenes, under the measures of Section 2.6 and against the overlap-aware reference the generator recorded when it drew them. Table 1 reports every measure overall and by density stratum; the paragraphs below take the headline score, how it moves with density, and what that density label does and does not mean.

Mean common-fragment pairwise F_1 over the 128 held-out scenes was 0.85 (density-stratified 95% bootstrap CI [0.84, 0.86]); Table 1 gives precision, recall, the adjusted Rand index and both variation-of-information components, overall and by density. These scenes used unseen seeds from the same procedural generator as development data, so this is same-generator generalisation and not distribution transfer (Supplementary Section S6).

Performance decreased with scene density: mean F_1 fell from 0.91 [0.89, 0.93] at 20% coverage to 0.77 [0.76, 0.79] at 60%, monotonically through the intermediate strata. The same decline appears on the paired clean-against-degraded set below, where the clean and the degraded curves separate at low coverage and meet at 50–60%: degradation costs least exactly where the score is lowest, because there the tangles rather than the gaps set the difficulty. Per-scene runtime had median 1.12 s, ranging to 34.06 s on the densest scenes; Section 3.5 measures cost properly, where the transferable quantity is the scaling exponent rather than any absolute time.

Areal coverage labels these strata but does not cause the difficulty they report, and it cannot: PLECTA is handed a centreline mask and never a width. Rendering one geometry at three bundle widths was tested directly on 36 development scenes, and it leaves the input mask and the overlap-aware reference byte-identical while areal coverage varies by up to a factor of 2.49; F_1 moved across those renderings by 0.0000. What does move the score is centreline length density, mean F_1 falling from 0.92 to 0.84. A coverage stratum should therefore be read as a proxy for the length density accompanying it under this generator.

How much of that score the input mask fixes, rather than the grouping, is separable. On the separate 50-scene controlled set, mean F_1 was 0.82 from degraded axes and 0.84 from the corresponding clean axes. The paired clean-minus-degraded change was 0.0224 (95% CI [0.0113,

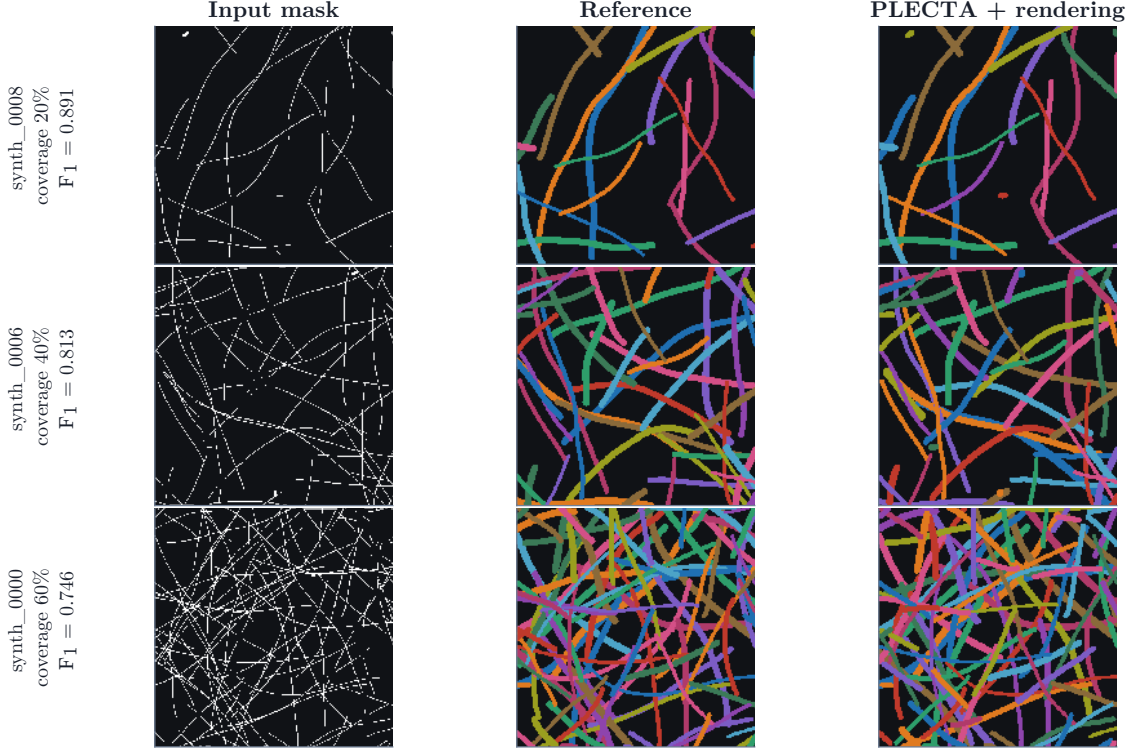


Figure 7. Reconstruction at three coverage levels, each the scene closest to the median F_1 of its stratum, shown as a horizontal window of the scene. Columns: the degraded input mask, the overlap-aware reference, and PLECTA’s strands at their fitted widths. The score beside each row is the grouping’s, not the rendering’s. Strand colours are arbitrary and not matched between columns.

0.0333]). Modelled gaps and raster irregularities therefore caused a small resolved loss under this generator.

Figure 7 shows what the score corresponds to scene by scene, taking at three coverage levels the scene whose F_1 is closest to the median, so the rows are typical rather than selected.

3.2 Depth decomposition

The stages above reconstruct strands in projection. A crossing pixel is assigned to both strands that own it, but nothing so far says which of the two lies on top. The depth stage answers that, reading the greyscale after the grouping is fixed and leaving every strand exactly as the stages above left it. Its evidence is the continuity of each strand’s own brightness across the crossing, weighed against the noise measured there; from those local calls it imposes one globally consistent vertical order and places each strand at a height at which nothing interpenetrates.

It is evaluated on 12 scenes from the depth-resolved generator of Section 2.2, at two areal coverages, in two input conditions. Given the reference 2-D geometry, the depth decision is measured alone; given only the binary axis mask, the same stage runs on PLECTA’s own reconstruction and carries its error. Table 2 reports both, under the measures Supplementary Section S5 defines. Its two blocks are not interchangeable. The crossing rows score the over/under call at each crossing, one score per crossing, taken before anything is reconciled; the stacking rows score the height field the solver returns once every call has been pooled, one score per pair of strands. Two strands that cross three times are three crossings and one pair.

The stage is accurate where it has evidence and blind where it has none, and the table separates the two. At a crossing it decides, order accuracy runs 0.88 to 0.96 whether the 2-D geometry was given or reconstructed and whether the scene carries the sparser crossing load or

Table 2. The depth stage on 12 synthetic scenes whose true vertical order is known. Chance is 0.5 for every accuracy shown.

Areal coverage	Given 2-D geometry		Reconstructed 2-D	
	20 %	40 %	20 %	40 %
<i>At each crossing</i>				
Projected crossings per scene	28	150	28	150
Crossings recovered	1.00	1.00	0.86	0.75
Order accuracy, all crossings	0.81	0.74	0.73	0.56
restricted to those it decided	0.89	0.88	0.96	0.91
Declined, of the crossings it found	0.08	0.16	0.12	0.18
<i>In the stacking it outputs</i>				
Depth order, all strand pairs	0.67	0.71	0.70	0.72
restricted to pairs that cross	0.86	0.86	0.91	0.85
Pairs within one crossing component	0.39	0.84	0.36	0.75
Layers in the reference	5.2	8.7	5.2	8.7
Layers recovered	5.2	8.3	5.3	8.0
Layer agreement	0.66	0.42	0.70	0.38

roughly five times it. That is what the decision is worth wherever the image speaks to a pair of strands. How often it does is the rest of the table.

Over every pair of strands in the scene, order accuracy falls to 0.70, and that is an absence of evidence rather than a failure of the decision. Depth propagates only along crossings, so two strands that never meet, and are joined by no chain of strands that do, are related by nothing the image contains; their order is a stated tie-break, and no method could do better on them. Only 0.36 to 0.84 of strand pairs lie within one connected component of the crossing graph, which is the fraction for which any evidence exists at all. That alone accounts for the collapse. Suppose the stage orders the connected pairs at the rate it achieves on pairs that cross and guesses the rest: at 20 % coverage the two terms come to 0.65, against 0.70 measured. At the sparser coverage the stage beats that model, so the unconnected pairs are evidently not quite coin flips; at the denser end it falls short, which is what error accumulating along the long transitive chains of a large component would do. Depth claims are therefore scoped to the connected component, and a globally ordered film would need evidence that does not travel through crossings, such as defocus, or a second view, rather than a smoothness assumption, which these scenes would not support: their heights are drawn independently, so proximity carries no depth information by construction.

Two more gaps sit inside the crossing rows themselves, and neither is the shortage of evidence just described: at a crossing the two strands do meet, so something in the image bears on their order. The all-crossings row is lower than the decided row because it charges the stage for crossings the 2-D stage never recovered and for crossings it declined, and the table gives each of those its own rate. That is also why the decided row must be read beside them: what the stage declines are the hard crossings, so a higher decided accuracy can be a stage answering less often rather than one deciding better.

Layer agreement degrades the way the global order does, falling from 0.70 to 0.38 as coverage doubles while the recovered layer count stays close to the reference: the ordering is locally right, so the number of layers is about right, while small order errors move strands across layer boundaries.

Figure 8 shows what the stage produces on one synthetic scene, from the binary axis mask through to the film the reconstruction implies; Figure 9 is the same sequence on a real CNT SEM field, which no depth ground truth can score.

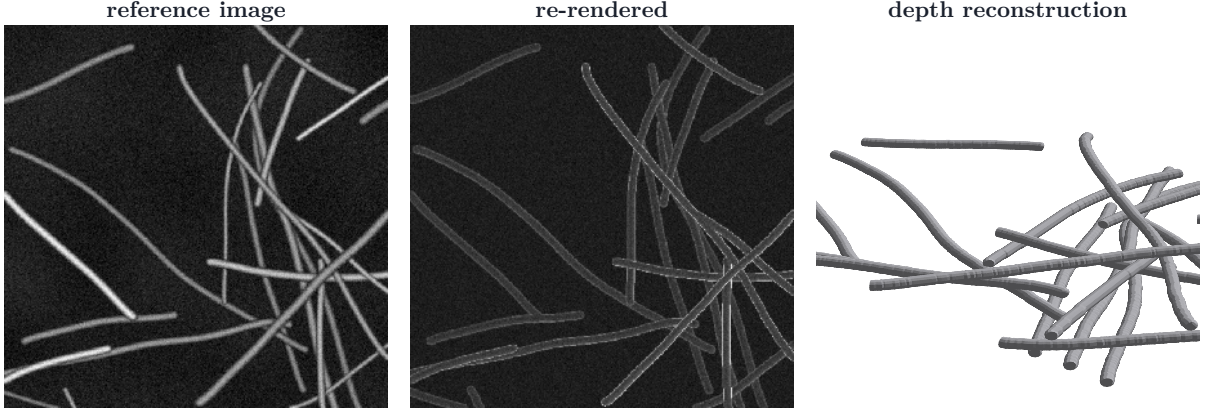


Figure 8. The depth stage on one synthetic scene, reconstructed from the binary axis mask. The image the scene was rendered from, that scene re-rendered from the reconstruction through a forward SEM model, and the film the reconstruction implies. The re-rendered panel is a consistency check on depths and radii, not a scored quantity: the difference Table 2 reports between input conditions is a difference in which crossings were *found*, which a rendered image cannot show.

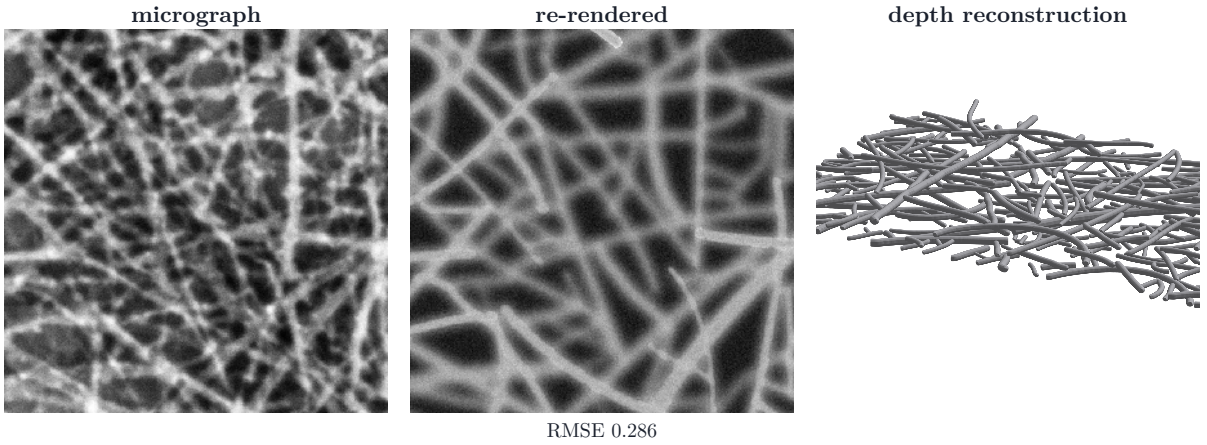


Figure 9. The same comparison on a real CNT SEM field, reconstructed from the axis of annotator A’s annotation. Nothing here is scored: real SEM has no depth ground truth, since the annotation records which pixels belong to which strand, not which lies on top. It shows that the stage runs end to end and re-renders to something recognisable. The forward model’s appearance parameters are fitted to this micrograph, so the residual beneath the panel is in-sample.

3.3 Proof of concept on real SEM

PLECTA works on real imaging when a second technique supplies the mask, and the upstream segmenter is such a technique.

The upstream segmenter (Figure 10) ends in a binary strand-axis mask [42], and PLECTA reads that mask and nothing else. Three manually annotated CNT-sheet fields separate the two: PLECTA is run once on the skeleton of annotator A’s annotation and once on the nnU-Net axis, with annotator A’s annotation as the reference in both cases, so the two conditions differ only in the mask. The first is an oracle-like input condition rather than an end-to-end result, asking what the grouping can do when the axis evidence is curated; only the second says what an automatic mask allows. Figure 11 shows both on one field, whole frame, with each input axis beside what PLECTA made of it.

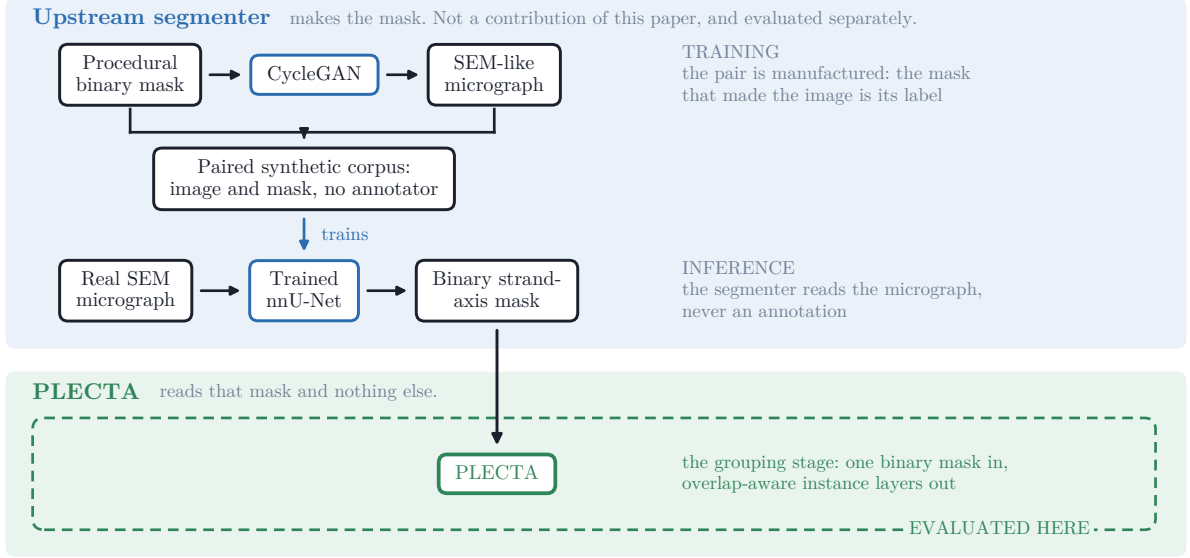


Figure 10. Where the mask comes from. The segmenter trains on manufactured pairs, so no micrograph is ever hand-annotated, and the binary axis mask it produces is the whole interface to PLECTA. It is not a contribution of this paper and is not evaluated here; it carries its own evaluation [42], and only the dashed box is measured.

Table 3. Reconstruction from the manual-derived and the nnU-Net axis in three CNT SEM fields, under the measures of Section 2.6. *Coverage* is the fraction of the frame the full-width reference strands cover (Section 2.2), not the thin axis mask’s. Detection F_1 is at IoU 0.1, tolerance 2 px.

Field	Coverage	Input axis	F_1	Join F_1	Det. F_1
B58_100	44.0 %	Manual-derived	0.89	0.93	0.94
		U-Net	0.46	0.56	0.63
B58_110	46.7 %	Manual-derived	0.84	0.91	0.93
		U-Net	0.45	0.59	0.60
B58_300	41.5 %	Manual-derived	0.95	0.96	0.96
		U-Net	0.47	0.62	0.58

What the three fields say, per field and per condition, is Table 3. Averaged over them, PLECTA reaches $F_1=0.89$ from the manual-derived axis and 0.46 from the nnU-Net axis. The first figure is what this section exists to establish: the grouping holds on axis masks read off real micrographs and not only on the generator’s. The second is a different claim and a weaker one, and the paragraphs below take the two apart; what it is worth is settled in Section 4, where it is read against what a second human annotator achieves on these same fields, which is less. 3 fields cannot estimate a population, and the inferential unit is the field.

Most of the distance between the two conditions is the mask, and what the method loses to it is joins it declines rather than strands it fuses. Beside annotator A’s axis, the nnU-Net axis is a second, independent reading of the same strands, and it arrives thicker, broken into more pieces and displaced by a pixel or two almost everywhere, which panels (b) and (d) of Figure 11 show side by side. Fragmentation is what that costs, and the pairwise score charges for it twice over: being quadratic in fragments, a strand the mask breaks into several pieces is charged once for every pair of pieces it fails to rejoin. Join F_1 , defined in Supplementary Section S3, charges each break the mask brings within rejoining reach once instead, and under it the gap narrows without closing. What is left is one-sided: on B58_110’s nnU-Net axis a join precision of 0.68 against a recall of 0.53 is a method leaving strands apart rather than running them together, which is what panel (e) of Figure 11 shows.

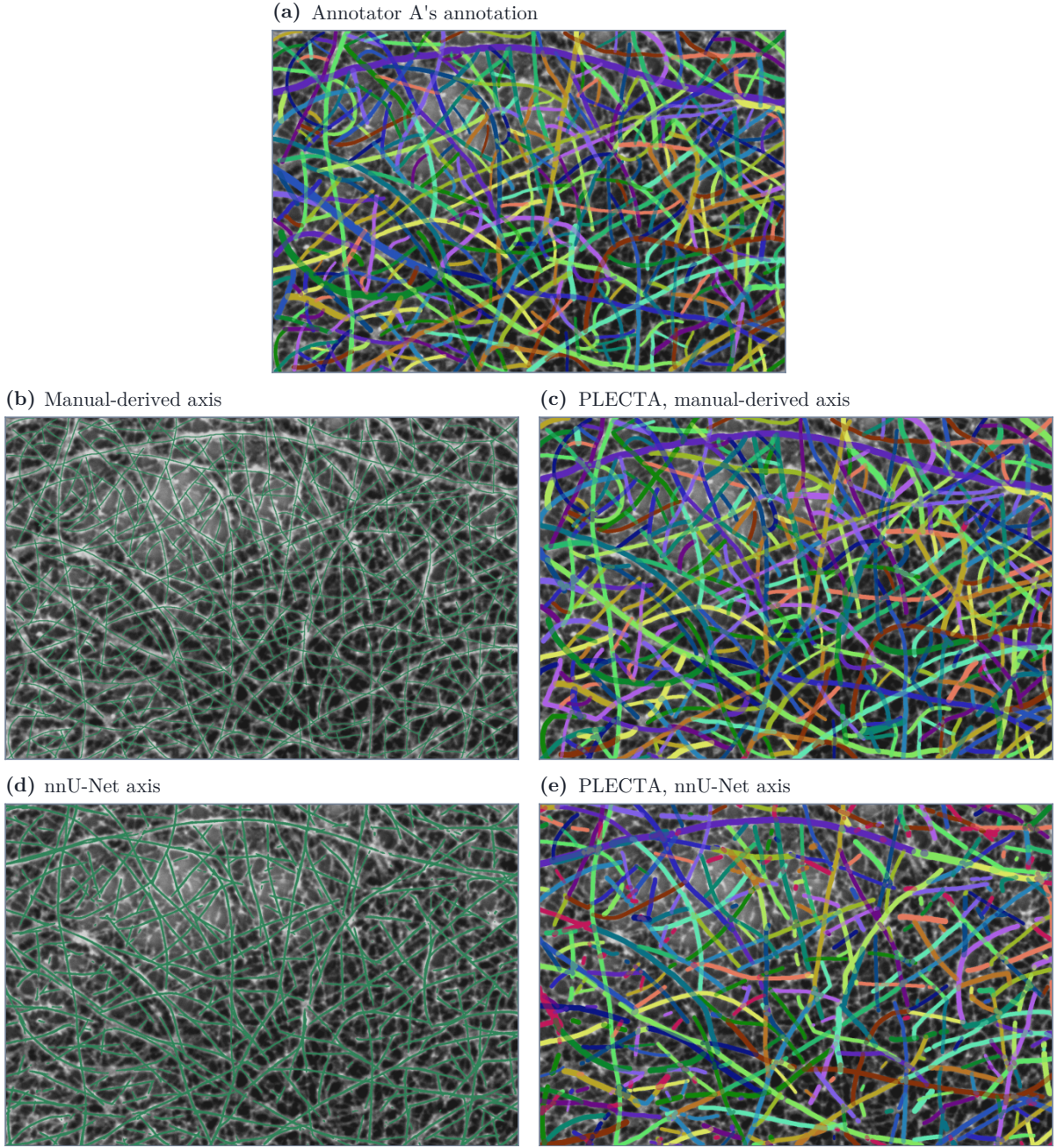


Figure 11. One CNT SEM field, B58_110, whole frame, a row per mask source. (a) annotator A’s annotation; (b) its skeleton, the manual-derived axis, and (c) PLECTA from it; (d) the segmenter’s axis, and (e) PLECTA from that, same parameters. (b) and (d) are the input masks, in one colour, dilated a pixel for legibility; the rest are at width, through the layer of Section 2.5.1, which runs after the grouping is fixed and moves no pixel between strands. A predicted strand takes the colour of the annotated strand it matched; crimson matched none.

How much of that remainder is the segmenter’s, and how much is the reference’s, this material cannot separate. Given annotator B’s skeleton in place of the network’s, PLECTA scores 0.38 against annotator A, below what the network’s axis reaches on every field and inside the 0.36–0.53 at which the two annotators agree with no method in the loop. A second human reading of these strands is therefore no better an axis than the segmenter’s, and the nnU-Net figure is bounded by how far one annotator’s definition of a strand can be reproduced at all, which Section 4 takes apart.

Table 4. Component ablations and the two controls, on development scenes. The three blocks use different scene sets, so n is given per row and each ΔF_1 is against PLECTA on that row’s own scenes; the column is not comparable across rows of different n .

Variant	n	F_1	ΔF_1	ARI
Fixed configuration	84	0.87	+0.000	0.86
No joint junction solve	30	0.78	-0.080	0.77
No gap bridging	84	0.73	-0.139	0.72
Single round	84	0.85	-0.020	0.84
No chain-extended frames	84	0.85	-0.021	0.84
No round schedule	84	0.86	-0.004	0.86
No curvature term	84	0.86	-0.005	0.86
No short-connector node merging	84	0.85	-0.022	0.84
No junction-cluster consolidation	84	0.80	-0.063	0.80
No junction chord-turn term	84	0.68	-0.186	0.67
Whole linker replaced	50	0.68	-0.145	0.67

3.4 Development ablations

The sections above measure what PLECTA does. This one asks which of its parts the score depends on, by removing one at a time and rescoring, and which two of them are not parts of it at all but controls. Everything here is measured on the development scenes and none of it reuses the held-out set, so the numbers below are diagnostic and are not comparable with the held-out result.

What each part is worth. Table 4 varies one thing at a time. Two rows change what a link costs, dropping the junction chord-turn term or the curvature term; two change the graph the linker reads, leaving neighbouring junction clusters unconsolidated or arms too short to merge the crossings they join; two change the iteration, running a single round or removing the schedule that relaxes the gates; and one removes gap bridging, so identity never crosses a break in the mask. The three largest losses are the junction chord-turn term, gap bridging and junction-cluster consolidation.

Two rows are controls rather than components. Removing the joint solve leaves PLECTA otherwise intact and pairs each node’s arms cheapest edge first, rather than by whichever complete pairing of that node is cheapest overall; a cheap edge taken early can then strand two arms that would have paired well together, which is why that arm fails by division rather than by merging. It also runs at parameters that were selected with the exact matcher in the loop, a bias in the exact matcher’s favour. The last row replaces the linker outright, skeletonising the same mask and greedily pairing the branch ends that turn least inside fixed distance and angle gates, which is the pairing principle shared by SIFNE, GraFT’s path cover and Wegmayr et al. [10, 34, 35].

3.5 Published methods on identical masks

The sections above measured PLECTA on its own. This one places it against established methods for the same task, on two bodies of material: the 50 synthetic scenes the comparison was designed on, and the three annotated CNT SEM fields of Section 3.3, where the manual-derived axis lets the same comparison run on real imaging. Four published methods are scored beside PLECTA on identical masks, through the same conversion and metric code, over scenes generated after every configuration was fixed. What is compared is narrow: the overlap-aware grouping of a binary axis mask of dense, crossing strands, at the coverages and tangling of the scenes described in Section 2.2. Each of these methods was built for a different setting, and how it performs in that setting is not what these numbers measure. DNAi [9], GraFT [10] and SIFNE [34] are run from their own released sources. For the fourth, Basu et al. [39], no implementation was found,

and it is reimplemented here. Running someone else’s method on data it was not built for takes decisions, and each of those decisions is stated here.

Table 5. PLECTA against four published methods, identical masks and one scorer, ordered by F_1 . The upper block is the degraded synthetic scenes; the lower is the annotated CNT SEM fields of Section 3.3 on the manual-derived axis, a plain mean over the three. Variation of information in bits, lower better; higher better for the rest.

* our reimplementation; no implementation of that paper was found. Supplementary Section S4 details it.

Method	F_1	Precision	Recall	ARI	VI total ↓	Crossing	Median s/scene	
						fidelity	at 20 %	at 60 %
<i>Synthetic scenes, degraded axis</i>								
PLECTA	0.82	0.83	0.81	0.82	0.57	0.81	0.14	4.09
SIFNE	0.71	0.77	0.67	0.70	0.93	0.77	3.42	5.68
Basu*	0.54	0.66	0.47	0.53	1.37	0.76	2.56	54.20
GraFT	0.49	0.67	0.40	0.48	1.56	0.68	2.44	29.74
DNAi	0.34	0.46	0.28	0.33	2.11	0.58	0.04	0.56
<i>Real CNT SEM fields, manual-derived axis</i>								
PLECTA	0.89	0.86	0.93	0.89	0.33	0.93		
SIFNE	0.87	0.86	0.88	0.86	0.37	0.92		
Basu*	0.76	0.74	0.78	0.76	0.59	0.89		

How each was run. Each released method has a front end that precedes the stage assigning identity, and in each case the mask is injected where that stage begins, so grouping is not confounded with segmentation and every stage that assigns identity is their own code. Each was also run at a configuration selected on the same 84 development scenes PLECTA was configured on rather than at its own defaults. Every one of these adaptations favours the comparator, and by a wide margin: run end to end on the greyscale, GraFT scores 0.10 against 0.49 injected, and the re-selected configurations are worth 0.19 to 0.34 for DNAi and 0.55 to 0.71 for SIFNE. Both are reported. Supplementary Section S4 gives each injection point, each selection, what was not attempted, and how each method fails.

Favoured in all those ways, the four order as Table 5 shows, and hold that order at every density and on both mask conditions (Figure 12). SIFNE is the strongest of them, so the margin rests on a method configured for this data rather than on a default that suits it badly. Two entries read against that ordering. The first is DNAi, whose deficit is a limitation of domain and density rather than a general ranking: it is built for DNA-fibre micrographs, which are sparse, and where the network is sparse and the axis clean it recovers 0.66 at 20% coverage. The second is Basu et al. [39], the closest structural relative PLECTA’s junction matching has in that it too optimises over a neighbourhood and prices joins on curvature. It stands second only to PLECTA on Crossing Fidelity while its pairwise F_1 sits far below both released methods, so the curvature-priced matching does real work at the junctions and what costs it is everything around that decision. That is why the priced refusal of Section 2.4 is claimed as part of a combination rather than on its own. Supplementary Section S4 gives both in full.

Each method on its own ground. That ordering is drawn from one generator, so it was tested in the regimes the two released methods were built for. On GraFT’s own generator, run unmodified at its published settings over 40 scenes in 8 density strata, PLECTA leads at every density and nothing crosses over, including inside the range GraFT itself published, and the field-standard measures order the three identically. SIFNE released no generator, so the sparse cobweb topology its paper validates against was rebuilt here from that description; on that reconstruction the two are level where the paper works, 0.97 against 0.97, and part only slightly at the denser end. That places the difference in how tangled the crossings are rather

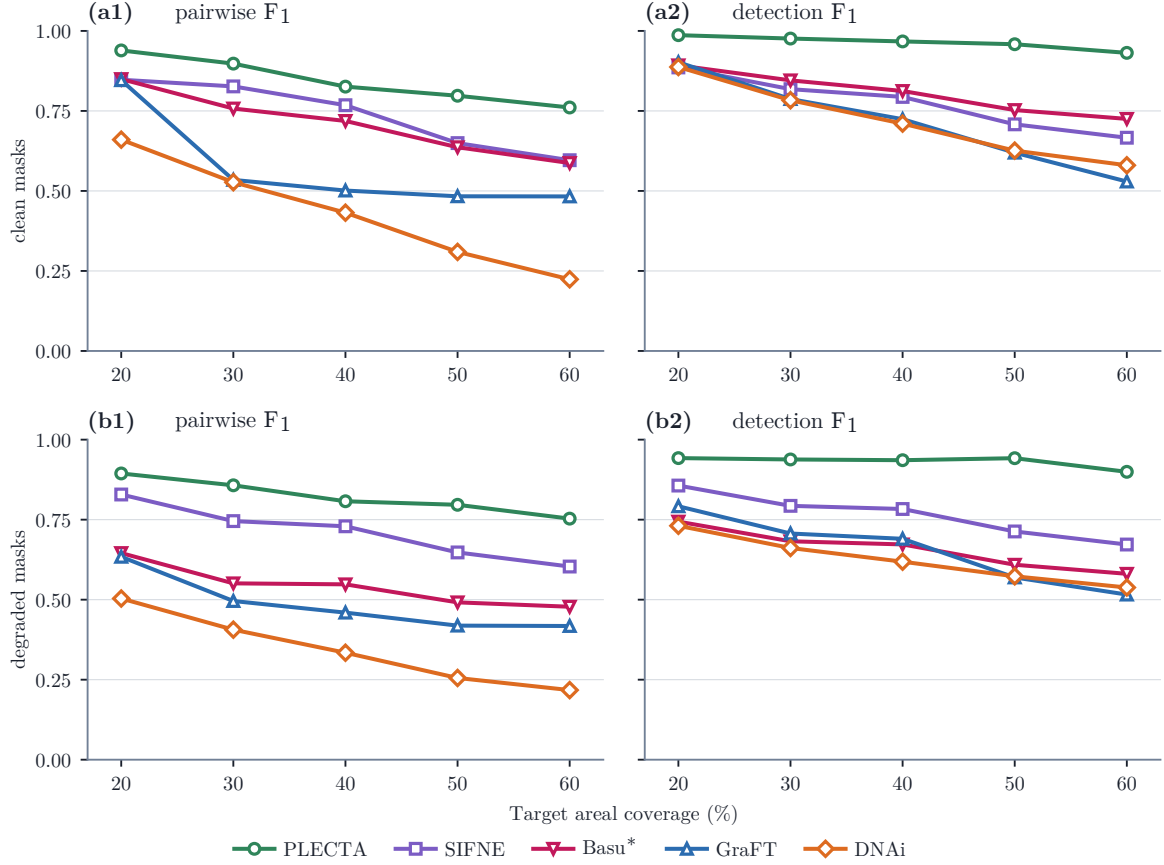


Figure 12. Five methods on identical masks, by coverage stratum: (a) the undegraded axis, (b) the degraded one; columns are pairwise F_1 , which counts pairs of fragments, and detection F_1 at IoU 0.1 with a 2 px tolerance, which counts objects (Section 2.6). The two disagree where a method fragments, since only the first is quadratic in fragmentation. Crossing Fidelity is in Table 5. Means of ten scenes. * our reimplement; none was found.

than in how much strand the frame holds, though a topology we rebuilt is weaker evidence than a generator its authors released. Supplementary Section S4 gives both density series in full.

Cost. Accuracy is not the only axis on which these methods differ. Per-scene cost was measured warm, single-threaded and pinned to one core, timing each method’s own computation only (Table 5; 3 GraFT scenes ran past the 600 s budget and were stopped rather than timed). These are independently optimised codebases, so absolute times measure implementation as much as algorithm; the transferable quantity is scaling, and the log-log exponents against foreground pixels are 2.11, 2.65 and at least 2.01. PLECTA has the largest exponent among the methods timed to completion; GraFT’s is a lower bound, because the scenes missing from its fit are the slowest ones, and only GraFT is heavy-tailed.

On real imaging. The two comparators that read a binary axis were run over the three annotated fields as well, on the manual-derived masks of Section 3.3 and through the scoring path that reproduces PLECTA’s stored numbers for those fields exactly. Averaged over the three, PLECTA reaches $F_1=0.89$ against 0.87 for SIFNE and 0.76 for Basu, and leads on every measure reported. The ordering established on synthetic scenes therefore holds on masks read off real micrographs.

It survives by much less, and the input is the likely reason. A skeleton of a hand annotation is about as clean an axis as a real field can supply: unbroken, thin, and placed where a person judged the strand to be. What separates these methods on the synthetic rows is what each does

with evidence that is fragmented and displaced, and this condition supplies almost none of it, so it compresses exactly the differences the comparison is for.

4 Conclusions

PLECTA reconstructs overlapping strand instances from a binary strand-axis mask and nothing else. It solves every junction, of any degree, as a single exact maximum-weight partial matching in which leaving an arm unmatched carries an explicit price, so a strand is allowed to end rather than have a continuation invented for it, and a crossing pixel is emitted to every strand that claims it.

- Over 128 held-out synthetic scenes from unseen seeds it reaches mean pairwise $F_1=0.85$ [0.84, 0.86], falling from 0.91 to 0.77 as areal coverage rises from 20 to 60 %.
- On identical masks it leads four published methods at every density, and the ordering holds on a competitor’s own generator, run unmodified at its published settings.
- What carries the result is geometric: the largest ablation losses are the junction chord-turn term and gap bridging.
- A secondary stage reads the greyscale to order strands in depth. It is correct on 0.88–0.96 of the crossings it commits to, against a chance level of 0.5, and abstains where the image does not separate the pair by more than the local noise.
- The grouping core uses no random number generator and no training data, and its configuration is published, so one mask yields one answer on every run.
- The route from a greyscale micrograph to those instances needs no hand-annotated image at any point. The upstream segmenter is trained on manufactured image and mask pairs, in which the mask that made the image is its own label, so the demonstration runs end to end on real SEM without a single manually annotated micrograph. It is not training-free, but the training it uses costs no annotation.

Bounds.

- Within the generator, the score does not turn on how the strands are shaped: sweeping its bending stiffness over 50–150 % of default, and its curve wavelength over 10–40 px, moves the level means of F_1 by at most 0.05 and 0.08, against a scene-to-scene scatter of 0.05 within a level (Supplementary Section S6).
- The comparison runs four methods on data none of them was built for. It supports a mechanical claim: that pairing endpoints greedily, or committing with no option to decline, loses identity in dense crossings.
- The remaining sensitivity is the segmenter’s. PLECTA can bridge a short omission, but it cannot infer a strand the mask omits, nor reliably reject a false axis introduced upstream.
- The real-material demonstration is a case study of three CNT SEM fields and cannot estimate generalisation. Scored against the annotator whose skeleton it was given, PLECTA reaches 0.79–0.95, and across annotators 0.37–0.51, while annotators A and B agree at 0.36–0.53 with no method in the loop. The disagreement is systematic rather than error: B exports 1.85–2.83 times as many objects as A, shorter and far more fragmented, which is two answers to what an instance *is*. A real-data score is therefore bounded by that disagreement rather than by the method, and identity through an occlusion is the part annotation cannot settle.
- Applicability follows the premise. The evidence is directional, so where direction does not survive a crossing, as in freely flexible chains, the method fails rather than degrades.

What would move any of this is independent real fields sealed before configuration, validated physical widths, further released implementations run in a matched comparison, and an external-generator benchmark with sealed seeds.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The PLECTA source, complete fixed configuration, compact seed manifest, and aggregate machine-readable evidence are available at <https://github.com/Jorallan/PLECTA>. A graphical application that runs the same reconstruction on a supplied mask, for readers who would rather not call the library directly, is at https://github.com/Jorallan/PLECTA_APP. The annotation tool of Section 2.3 is at <https://github.com/Jorallan/ImageAnnotator>; it is listed because the reference is machine-assisted rather than freehand, so the ridge snapping, spline re-rendering and full-width-at-half-maximum widths it applies are part of how the reference every real-field score is taken against was made. Generated scenes, detailed per-scene records, and development diagnostics are retained locally.

Acknowledgements

The authors thank annotator B, whose independent re-annotation of 3 of the fields in Section 2.3 is what the reader-agreement bound in Section 4 rests on. The software used is NumPy, SciPy, scikit-image, NetworkX and Matplotlib, and the authors thank their maintainers.

References

- [1] Yuezhou Wang, Igor Ostanin, Cristian Gaidau, and Traian Dumitrică. Twisting carbon nanotube ropes with the mesoscopic distinct element method: Geometry, packing, and nanomechanics. *Langmuir*, 31(45): 12323–12327, 2015. doi: 10.1021/acs.langmuir.5b03208.
- [2] Michael F. L. De Volder, Sameh H. Tawfick, Ray H. Baughman, and A. John Hart. Carbon nanotubes: Present and future commercial applications. *Science*, 339(6119):535–539, 2013. doi: 10.1126/science.1222453.
- [3] Jonathan N. Coleman, Umar Khan, Werner J. Blau, and Yuri K. Gun’ko. Small but strong: A review of the mechanical properties of carbon nanotube-polymer composites. *Carbon*, 44(9):1624–1652, 2006. doi: 10.1016/j.carbon.2006.02.038.
- [4] Alexey N. Volkov and Leonid V. Zhigilei. Structural stability of carbon nanotube films: The role of bending buckling. *ACS Nano*, 4(10):6187–6195, 2010. doi: 10.1021/nn1015902.
- [5] Igor Ostanin, Roberto Ballarini, David Potyondy, and Traian Dumitrică. A distinct element method for large scale simulations of carbon nanotube assemblies. *Journal of the Mechanics and Physics of Solids*, 61(3): 762–782, 2013. doi: 10.1016/j.jmps.2012.10.016.
- [6] Igor Ostanin, Traian Dumitrică, Sebastian Eibl, and Ulrich Rude. Size-independent mechanical response of ultrathin carbon nanotube films in mesoscopic distinct element method simulations. *Journal of Applied Mechanics*, 86(12):121006, 2019. doi: 10.1115/1.4044413.
- [7] Bernard K. Wittmaack, Abu Horaira Banna, Alexey N. Volkov, and Leonid V. Zhigilei. Mesoscopic modeling of structural self-organization of carbon nanotubes into vertically aligned networks of nanotube bundles. *Carbon*, 130:69–86, 2018. doi: 10.1016/j.carbon.2017.12.078.
- [8] Oday Allan, Tanmaya Mishra, Robert Jan Meijer, and Matthijn de Rooij. Micro-CT analysis of geometrical distribution of filaments in double-twisted three-strand ropes: Comparison with analytical model. *Journal of Industrial Textiles*, 55:15280837251333116, 2025. doi: 10.1177/15280837251333116.
- [9] Clément Ployout, Yosra Mehrjoo, Renaud Duval, Marie Carole Boucher, Santiago Costantino, and Hugo Wurtele. DNAi: An open-source AI tool for unbiased DNA fiber analysis. *Nucleic Acids Research*, 54(7): gkag335, 2026. doi: 10.1093/nar/gkag335.

- [10] Isabella Østerlund, Arne Neumann, Zhiming Ma, Yansong Miao, Staffan Persson, and Zoran Nikoloski. GraFT: A robust network-based spatiotemporal analysis of filamentous structures. *Science Advances*, 12(13): ead4132, 2026. doi: 10.1126/sciadv.adz4132.
- [11] Liyu Dong, Haibin Hang, Jin Gyu Park, Washington Mio, and Richard Liang. Detecting carbon nanotube orientation with topological analysis of scanning electron micrographs. *Nanomaterials*, 12(8):1251, 2022. doi: 10.3390/nano12081251.
- [12] Tamjid Imtiaz, Jacques Doumani, Fuyang Tay, Natsumi Komatsu, Stephen Marcon, Motonori Nakamura, Saunab Ghosh, Andrey Baydin, Junichiro Kono, and Ahmed Zubair. Facile alignment estimation in carbon nanotube films using image processing. *Signal Processing*, 202:108751, 2023. doi: 10.1016/j.sigpro.2022.108751.
- [13] Sahar Beigzadeh and Jeffrey E. Shield. Utilizing local orientation image analysis for microstructure quantification in additive manufacturing. *Materials Characterization*, 210:113761, 2024. doi: 10.1016/j.matchar.2024.113761.
- [14] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, volume 9351 of *Lecture Notes in Computer Science*, pages 234–241, 2015. doi: 10.1007/978-3-319-24574-4_28.
- [15] Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18:203–211, 2021. doi: 10.1038/s41592-020-01008-z.
- [16] Tiberiu Stan, Zachary T. Thompson, and Peter W. Voorhees. Optimizing convolutional neural networks to perform semantic segmentation on large materials imaging datasets: X-ray tomography and serial sectioning. *Materials Characterization*, 160:110119, 2020. doi: 10.1016/j.matchar.2020.110119.
- [17] Tsimur Davydzenka, Daniel Sinclair, Nikhilesh Chawla, and Pejman Tahmasebi. Deep-layers-assisted machine learning for accurate image segmentation of complex materials. *Materials Characterization*, 192:112175, 2022. doi: 10.1016/j.matchar.2022.112175.
- [18] María Celeste Ramírez Trujillo, Teresa E. Alarcón, Oscar S. Dalmau, and Adalberto Zamudio Ojeda. Segmentation of carbon nanotube images through an artificial neural network. *Soft Computing*, 21(3):611–625, 2017. doi: 10.1007/s00500-016-2426-1.
- [19] Nguyen P. Nguyen, Ramakrishna Surya, Matthew Maschmann, Prasad Callyam, Kannappan Palaniappan, and Filiz Bunyak. Self-supervised orientation-guided deep network for segmentation of carbon nanotubes in SEM imagery. In *Computer Vision – ECCV 2022 Workshops*, *Lecture Notes in Computer Science*, pages 412–428, 2023. doi: 10.1007/978-3-031-25085-9_24.
- [20] Ivânia Maria Lucinda De Donato, Juliana do Nascimento Lunz, Fernanda Davi Marques, Bráulio Soares Archanjo, Francisco José Pereira Lopes, and Gilson Antônio Giraldo. Deep learning-based segmentation of nanomaterial images acquired via scanning electron microscopy. *Revista de Informática Teórica e Aplicada*, 32(1):83–90, 2025. doi: 10.22456/2175-2745.143516.
- [21] Shuangmei He, Jiahao Tu, Yanlin Zhu, Weifu Li, Liming Tan, and Zi Wang. Instance segmentation and microstructure characterization based on powder microscopic image data. *Materials Characterization*, 234: 116051, 2026. doi: 10.1016/j.matchar.2026.116051.
- [22] Carsen Stringer, Tim Wang, Michalis Michaelos, and Marius Pachitariu. Cellpose: A generalist algorithm for cellular segmentation. *Nature Methods*, 18:100–106, 2021. doi: 10.1038/s41592-020-01018-x.
- [23] Uwe Schmidt, Martin Weigert, Coleman Broadus, and Gene Myers. Cell detection with star-convex polygons. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*, *Lecture Notes in Computer Science*, pages 265–273, 2018. doi: 10.1007/978-3-030-00934-2_30.
- [24] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3992–4003, 2023. doi: 10.1109/ICCV51070.2023.00371.
- [25] Carsten Steger. An unbiased detector of curvilinear structures. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 20(2):113–125, 1998. doi: 10.1109/34.659930.
- [26] Alejandro F. Frangi, Wiro J. Niessen, Koen L. Vincken, and Max A. Viergever. Multiscale vessel enhancement filtering. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI’98*, volume 1496 of *Lecture Notes in Computer Science*, pages 130–137, 1998. doi: 10.1007/BFb0056195.
- [27] Yoshinobu Sato, Shin Nakajima, Nobuyuki Shiraga, Hideki Atsumi, Shigeyuki Yoshida, Thomas Koller, Guido Gerig, and Ron Kikinis. Three-dimensional multi-scale line filter for segmentation and visualization of curvilinear structures in medical images. *Medical Image Analysis*, 2(2):143–168, 1998. doi: 10.1016/S1361-8415(98)80009-1.
- [28] Andrew M. Stein, David A. Vader, Louise M. Jawerth, David A. Weitz, and Leonard M. Sander. An algorithm for extracting the network geometry of three-dimensional collagen gels. *Journal of Microscopy*, 232(3):463–475, 2008. doi: 10.1111/j.1365-2818.2008.02141.x.
- [29] Jeremy S. Bredfeldt, Yuming Liu, Carolyn A. Pehlke, Matthew W. Conklin, Joseph M. Szulcowski, David R. Inman, Patricia J. Keely, Robert D. Nowak, Thomas R. Mackie, and Kevin W. Eliceiri. Computational segmentation of collagen fibers from second-harmonic generation images of breast cancer. *Journal of Biomedical Optics*, 19(1):016007, 2014. doi: 10.1117/1.JBO.19.1.016007.

- [30] Ting Xu, Dimitrios Vavylonis, Feng-Ching Tsai, Gijsje H. Koenderink, Wei Nie, Eddy Yusuf, I-Ju Lee, Jian-Qiu Wu, and Xiaolei Huang. SOAX: A software for quantification of 3D biopolymer networks. *Scientific Reports*, 5:9081, 2015. doi: 10.1038/srep09081.
- [31] Daniel A. D. Flormann, Moritz Schu, Emmanuel Terriac, Divyendu Thalla, Lucina Kainka, Marcus Koch, Annica K. B. Gad, and Franziska Lautenschläger. A novel universal algorithm for filament network tracing and cytoskeleton analysis. *The FASEB Journal*, 35(5):e21582, 2021. doi: 10.1096/fj.202100048R.
- [32] Nathan A. Hotaling, Kapil Bharti, Haydn Kriel, and Carl G. Simon. DiameterJ: A validated open source nanofiber diameter measurement tool. *Biomaterials*, 61:327–338, 2015. doi: 10.1016/j.biomaterials.2015.05.015.
- [33] Dorian Depriester, Sabine Rolland du Roscoat, Laurent Orgéas, Christian Geindreau, Benjamin Levrard, and Florian Brémond. Individual fibre separation in 3D fibrous materials imaged by X-ray tomography. *Journal of Microscopy*, 286(3):220–239, 2022. doi: 10.1111/jmi.13096.
- [34] Zhen Zhang, Yukako Nishimura, and Pakorn Kanchanawong. Extracting microtubule networks from superresolution single-molecule localization microscopy data. *Molecular Biology of the Cell*, 28(2):333–345, 2017. doi: 10.1091/mbc.E16-06-0421.
- [35] Viktor Wegmayr, Aytunc Sahin, Björn Sæmundsson, and Joachim M. Buhmann. Instance segmentation for the quantification of microplastic fiber images. In *2020 IEEE Winter Conference on Applications of Computer Vision (WACV)*, pages 2199–2206, 2020. doi: 10.1109/WACV45572.2020.9093352.
- [36] Yi Liu, Abhishek Kolagunda, Wayne Treible, Alex Nedo, Jeffrey Caplan, and Chandra Kambhamettu. Intersection to overpass: Instance segmentation on filamentous structures with an orientation-aware neural network and terminus pairing algorithm. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 125–133, 2019. doi: 10.1109/CVPRW.2019.00021.
- [37] Max Frei and Frank Einar Kruis. FibeR-CNN: Expanding Mask R-CNN to improve image-based fiber analysis. *Powder Technology*, 377:974–991, 2021. doi: 10.1016/j.powtec.2020.08.034. Published online 2020.
- [38] Yi Liu, Su Peng, Jeffrey Caplan, and Chandra Kambhamettu. Pick and trace: Instance segmentation for filamentous objects with a recurrent neural network. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023*, Lecture Notes in Computer Science, pages 635–645, 2023. doi: 10.1007/978-3-031-43993-3_61.
- [39] Saurav Basu, Chi Liu, and Gustavo Kunde Rohde. Extraction of individual filaments from 2D confocal microscopy images of flat cells. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 12(3):632–643, 2015. doi: 10.1109/TCBB.2014.2372783.
- [40] David Breuer and Zoran Nikoloski. DeFiNe: An optimisation-based method for robust disentangling of filamentous networks. *Scientific Reports*, 5:18267, 2015. doi: 10.1038/srep18267.
- [41] Jaydeep De, Li Cheng, Xiaowei Zhang, Feng Lin, Huiqi Li, Kok Haur Ong, Weimiao Yu, Yuanhong Yu, and Sohail Ahmed. A graph-theoretical approach for tracing filamentary structures in neuronal and retinal images. *IEEE Transactions on Medical Imaging*, 35(1):257–272, 2016. doi: 10.1109/TMI.2015.2465962.
- [42] Bastian Rühle, Julian Frederic Krumrey, and Vasile-Dan Hodoroba. Workflow towards automated segmentation of agglomerated, non-spherical particles from electron microscopy images using artificial neural networks. *Scientific Reports*, 11(1):4942, 2021. doi: 10.1038/s41598-021-84287-6.
- [43] Torben Peters, John Schumann, Kerstin Kämpf, and Asmus Meyer-Plath. AI-assisted recognition of fibre instances in scanning electron microscope images for aerosol characterization and exposure assessment. *Powder Technology*, 469:121909, 2026. doi: 10.1016/j.powtec.2025.121909.
- [44] Cian Gabbett, Luke Doolan, Kevin Synnatschke, Laura Gambini, Emmet Coleman, Adam G. Kelly, Shixin Liu, Eoin Caffrey, Jose Munuera, Catriona Murphy, Stefano Sanvito, Lewys Jones, and Jonathan N. Coleman. Quantitative analysis of printed nanostructured networks using high-resolution 3D FIB-SEM nanotomography. *Nature Communications*, 15:278, 2024. doi: 10.1038/s41467-023-44450-1.
- [45] Guillem Palou and Philippe Salembier. Monocular depth ordering using T-junctions and convexity occlusion cues. *IEEE Transactions on Image Processing*, 22(5):1926–1939, 2013. doi: 10.1109/TIP.2013.2240002.
- [46] Hyunmin Lee and Jaesik Park. Instance-wise occlusion and depth orders in natural scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21178–21189, 2022. doi: 10.1109/CVPR52688.2022.02053.
- [47] Menglei Chai, Lvdi Wang, Yanlin Weng, Yizhou Yu, Baining Guo, and Kun Zhou. Single-view hair modeling for portrait manipulation. *ACM Transactions on Graphics*, 31(4):116:1–116:8, 2012. doi: 10.1145/2185520.2185612.
- [48] T. Y. Zhang and C. Y. Suen. A fast parallel algorithm for thinning digital patterns. *Communications of the ACM*, 27(3):236–239, 1984. doi: 10.1145/357994.358023.
- [49] Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985. doi: 10.1007/BF01908075.
- [50] Marina Meilă. Comparing clusterings—an information based distance. *Journal of Multivariate Analysis*, 98(5):873–895, 2007. doi: 10.1016/j.jmva.2006.11.013.